

Raw test results [DB METRICS], [DB IO], and [DB IO DISK] is saved in Test3_result_endurance test_raw.txt, which was saved under the same directory as this report.

Endurance Test Report

Duration: 30 minutes. Target rate: 10,000 msg/sec (~80% of write throughput from test 2 stress test). Only the DB consumer was active; Redis broadcast was disabled to isolate DB persistence. Because the consumer writes to DB only and does not publish to Redis, totalReceived is 0 throughout.

Configuration

- Duration: 30 minutes
- Target rate: 10,000 msg/sec (~80% of write throughput measured in Test 2)
- Redis broadcast: disabled (DB persistence isolation)
- Queue buffer: 500,000

Key Findings

- Total messages sent: 16,815,950
- Total messages written to DB: ~12,986,433 (still draining at test end)

Results

The test exhibited three distinct phases.

Phase 1 - Steady State (t=0 min to t=2 min, 22:07:57–22:10:29)

Throughput matched the producer at 9,400 - 9,700 msg/sec. Queue depth stayed near 0. Latency: p50=7- 12 ms, p99=40 - 80 ms. The system handled sustained load without pressure. This phase ended when checkpointsTimed first incremented at 22:10:29, accompanied by a buffersCheckpoint spike to 1,160, marking the first forced PostgreSQL checkpoint.

Phase 2 - Periodic Checkpoint Degradation (t=2 min to t=18 min, 22:10:29–22:26:12)

A recurring pattern emerged every 2 - 3 minutes: throughput dropped to 6,000–8,000 msg/sec, queue depth spiked to 3,000 - 20,000, and p99 latency jumped to 700 - 2,000 ms. After each spike the DatabaseWriter overcorrected at 11,000 - 13,880 msg/sec to drain the backlog before recovering to steady state. Each event corresponds to a forced PostgreSQL checkpoint triggered when WAL volume exceeded max_wal_size, saturating the gp3 3,000 IOPS baseline.

Phase 3 - Irreversible Degradation (t=18 min to t=30 min, 22:26:12–22:37:57)

At 22:26:12 the queue hit the 500,000 buffer cap. Throughput collapsed to a sustained 4,000 - 6,500 msg/sec with periodic drops to 1,900 - 3,200 msg/sec during checkpoint events. Latency rose to p50=300 - 900 ms and p99=500 - 2,400 ms. The system never recovered. Messages arriving faster than the DB could process were dropped at the buffer.

Root Cause

Repeated forced checkpoints triggered by WAL volume exceeding max_wal_size=1 GB saturated the gp3 disk I/O budget. Each checkpoint flushed large volumes of dirty buffers (buffersCheckpoint spikes visible in DB IO DISK logs), stalling normal write operations. Checkpoint frequency increased as the database grew, preventing full recovery between events and ultimately causing irreversible queue accumulation.

Monitoring

Memory

Heap grew from 35 - 118 MB in Phase 1 to up to 548 MB by late Phase 3. This is not a memory leak - the growth correlates directly with the 500,000-message buffer filling under backpressure. There is a clear upward trend tied to queue accumulation, not random oscillation.

Connection pool

Active DB connections fluctuated 1 - 9 throughout. The HikariCP pool (size 12) was never exhausted and was not a factor in the observed degradation.

Disk space

Database size grew from ~23 MB at the start of active writing to ~5,549 MB (~5.5 GB) by end of test. No disk alarm triggered, but at this rate a longer test would exhaust available storage on the gp3 volume.

Lock wait time

Lock wait was 0 ms throughout. No lock contention observed at any phase.

Database Metrics

Queries per second

During active steady-state writing the DB IO logs show 98–132 transactions/sec. Each transaction is a batch insert of up to 500 messages, corresponding to ~9,400 msg/sec effective throughput - consistent with DB METRICS logs.

Buffer pool hit ratio

Ranged from 98.65 - 98.84% across the test. Nearly all data access was served from PostgreSQL shared buffer cache. readsPerSec was 0 for the entire test.

Disk I/O (DB IO DISK logs)

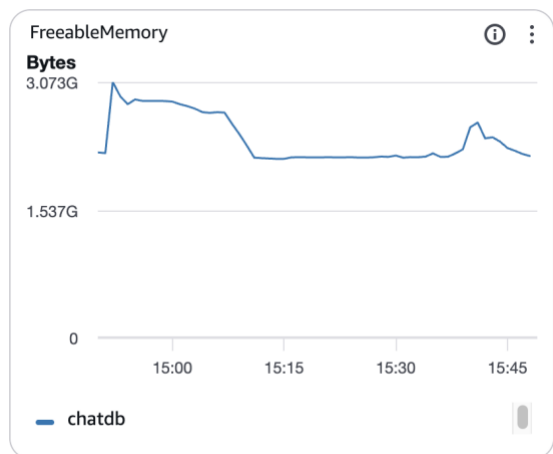
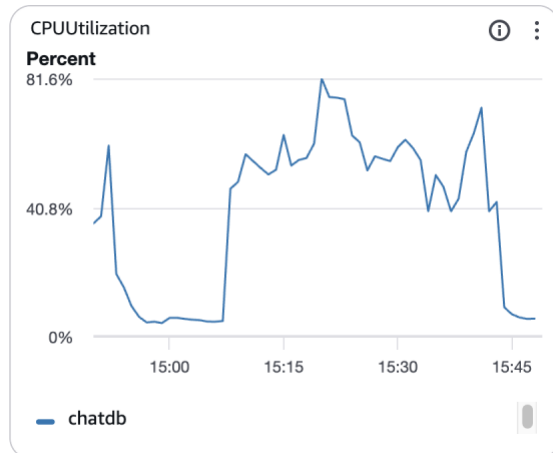
checkpointsReq was 3 at the start and remained 3 at the end of the test — it never incremented. The 3 required checkpoints were pre-existing. Forced checkpoint activity is evidenced by checkpointsTimed incrementing and buffersCheckpoint spiking, not by checkpointsReq.

backendWriteRatio reached 70–83% in Phase 3, indicating severe I/O saturation where user query threads were forced to perform writes normally handled by the background writer.

writeLatencyMs peaked at 21 ms during the worst checkpoint stalls, confirming disk I/O saturation. Normal operation: 0.01–0.15 ms.

Other Metrics from RDS:

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